

SMART INDIA HACKATHON 2026



SIH 2026 OFFICIAL IDEA SUBMISSION

- **Problem Statement ID** – 26143
- **Problem Statement Title** – Leveraging satellite imagery to determine Oil spills at sea along with AIS data correlations to identify vessel responsible for the spill.
- **Theme** – Disaster Management
- **PS Category** – Software
- **Team Name** – MarSlick
- **Idea Title** – MAR-SLICK: Autonomous Satellite Oil Spill Detection & AIS Vessel Attribution Platform

OPERATIONAL & DOMAIN METRICS

- **Primary Beneficiary:**
Indian Coast Guard (ICG) Maritime Rescue Coordination Centres (MRCC) & DG Shipping
- **Surveillance Scope:**
Indian EEZ (2.37 Million km²), Mumbai High, Gulf of Kutch, Bay of Bengal tanker corridors
- **Satellite Inputs:**
ESA Copernicus Sentinel-1 C-Band SAR (10m GRD), Sentinel-2 MSI Multi-Spectral Optical
- **Telemetry Data:**
ITU-R M.1371 Class-A AIS streams & MarineCadastre vessel archives
- **System Readiness:**
TRL-4 Working Prototype with 13/13 passing automated unit & integration tests

MAR-SLICK: AUTOMATED OIL SPILL ATTRIBUTION

Detailed Explanation of the Proposed Solution

- **Dual-Sensor Remote Sensing:** Ingests Sentinel-1 C-band SAR and optical imagery to detect dark oil slicks via radar backscatter damping.
- **Overlapping U-Net Tiling:** Splits large SAR scenes into 256x256 tiles with 32px overlap, stitching confidence masks without tile border seams.
- **Lagrangian Drift Hindcast:** Simulates backward-in-time drift using ocean surface currents and a 3% wind factor to pinpoint release origin.
- **Vectorized AIS Query:** Screens vessel traffic in a 50 km / 48 hr spatio-temporal cylinder of the origin in under 100 milliseconds.
- **Forensic Suspect Scoring:** Ranks suspect vessels combining spatial proximity, temporal delta, and navigational anomaly heuristics.

How It Addresses the Problem?

- **Solves Attribution Gap:** Bridges the historical gap between detecting an oil slick and identifying the culprit vessel responsible for illicit dumping.
- **Eliminates Manual Delays:** Replaces slow 48-72h manual GIS plotting with a sub-60-second automated forensic intelligence pipeline.
- **Corrects for Ocean Drift:** Accounts for wind and surface current displacement so ships are matched to release coordinates, not drifted positions.
- **Unmasks Dark Vessels:** Detects transponder shutoffs, loitering, and sudden speed drops used by polluters to obscure illicit discharge.
- **Court-Admissible Dossier:** Synthesizes GeoJSON polygons, drift tracks, and ranked MMSI profiles into evidence-grade reports for enforcement.

Innovation and Uniqueness of the Solution

- **Multi-Modal Closed Loop:** Seamlessly unifies SAR radar imagery, real-time hydrodynamic ocean models, and global AIS telemetry.
- **Sub-100ms Vectorization:** Replaces heavy spatial database joins with vectorized NumPy math, screening 10,000+ vessel pings in milliseconds.
- **Dynamic Met-Ocean Caching:** Live environmental data queries (Open-Meteo & CMEMS) with automated Redis caching and uncertainty cones.
- **Multi-Layer Safeguard Gates:** Out-of-Domain gate rejects optical non-SAR photos and Georeference gate enforces spatial anchoring.
- **Self-Contained Deployment:** Fully containerized architecture capable of air-gapped operation in coastal maritime rescue coordination centers.

1. SATELLITE SAR INGESTION & CALIBRATION

ESA Sentinel-1 C-Band SAR GRD (VV/VH dual-pol) & Optical
Radiometric calibration, speckle filtering, GeoTIFF spatial metadata

**2. TILING & U-NET SEMANTIC SEGMENTATION**

256x256 sliding tiles with 32px stride overlap (seamless stitching)
4-stage U-Net encoder-decoder (31M params) outputs binary mask (>0.5)

**3. VECTOR POLYGONIZATION & SPATIAL ANCHORING**

OpenCV contour extraction converts mask to valid WGS84 GeoJSON
Calculates geodesic area (sq km), perimeter, centroid (lat, lon)

**4. LAGRANGIAN MET-OCEAN DRIFT HINDCAST**

Numerical backward advection: $V_{drift} = V_{current} + 0.03 * V_{wind}$
Reconstructs spill origin coordinate (lat, lon) & expanding uncertainty radius

**5. SPATIO-TEMPORAL AIS TELEMETRY QUERY**

Searches vessel traffic in 50 km spatial radius & +/-48h time window
Vectorized NumPy Haversine distance screens 10,000+ pings in <100ms

**6. BEHAVIORAL ANOMALY & SUSPECT SCORING**

Detects AIS blackout gaps, sudden speed drops (>5kt), and loitering
Composite Score: $0.40 * Proximity + 0.35 * Temporal + 0.25 * Behavioral$

**7. OPERATIONAL GIS MAP & FORENSIC DOSSIER**

Interactive Leaflet GIS console with slick polygon, drift line, and tracks
Court-admissible evidentiary incident report for Indian Coast Guard & DG

ARCHITECTURE & CORE SUBSYSTEMS

- **SAR Vision Subsystem:** 4-stage U-Net CNN (31M params), 256x256 sliding tiles with 32px overlap, OpenCV contour-to-GeoJSON extraction.
- **Ocean Advection Subsystem:** Lagrangian backward trajectory advection: $V_{drift} = V_{current} + 0.03 * V_{wind}$ with expanding uncertainty envelope.
- **AIS Correlation Subsystem:** MarineCadastre CSV streaming parser, vectorized Haversine proximity screening, and kinematic anomaly detection (gaps, speed drops).
- **Command Console & APIs:** Django REST Framework, Celery asynchronous queue orchestrator, Leaflet.js mapping, and OpenAPI 3.0 documentation.

TECHNOLOGY STACK SPECIFICATIONS

- **Deep Learning & Vision:** PyTorch 2.0+, Torchvision, OpenCV, Albumentations, CUDA acceleration
- **Backend & Task Queue:** Python 3.12, Django 5.x, Django REST Framework, Celery 5.x, Redis 7.x
- **Hydrodynamics & Math:** Lagrangian Particle Engine, Open-Meteo Marine API, Copernicus CMEMS, NumPy
- **Geospatial & Telemetry:** Leaflet.js, PostGIS / SpatiaLite, Shapely, MarineCadastre AIS, NMEA 0183

ANALYSIS OF THE FEASIBILITY OF THE IDEA

- **Technical Feasibility:** Proven U-Net segmentation achieving <2s per tile; vectorized advection and scoring running in <100ms. 13/13 passing automated tests.
- **Operational Feasibility:** Direct ingestion of standard maritime telemetry (NMEA 0183 / MarineCadastre AIS) without requiring vessel transponder modifications.
- **Economic Viability:** Built on open-source frameworks and public European Space Agency Sentinel-1 satellite feeds, eliminating enterprise licensing fees.

POTENTIAL CHALLENGES AND RISKS

- **Look-Alike False Positives:** Natural biogenic slicks, low-wind calm ocean surfaces (<3 m/s), and algal blooms mimicking radar backscatter damping in SAR.
- **Met-Ocean Grid Resolution:** Gaps or coarse spatial resolution in real-time ocean current models near complex coastlines and shallow archipelagos.
- **Dark Vessels & AIS Blackouts:** Deliberate transponder shutdown or spoofed MMSI identity during illegal night-time bilge discharges.

STRATEGIES TO OVERCOME CHALLENGES

- **Multi-Feature SAR Rejection:** Texture analysis (GLCM), backscatter ratio profiling, and wind-speed threshold gating (>3 m/s) to reject look-alikes.
- **Dynamic Environmental Blending:** Hierarchical fallback between coastal HF-radar, Open-Meteo, and Copernicus CMEMS global currents with uncertainty cones.
- **Kinematic Gap & Anomaly Scoring:** Automated detection of AIS blackout gaps, sudden speed drops, and track extrapolation to unmask non-broadcasting vessels.

01

PHASE 1: BENCHMARK

Core Algorithm Validation

- Sentinel-1 SAR U-Net (31M params)
- Vectorized Haversine AIS (<100ms)
- 13/13 passing automated test suite

02

PHASE 2: PILOT TRIAL

Coastal Operational Testing

- Live Open-Meteo & CMEMS currents
- Indian Coast Guard MRCC trial
- Sub-60s automated alert dispatch

03

PHASE 3: FULL ROLLOUT

Pan-India EEZ Surveillance

- Full 2.37M km² coastal EEZ coverage
- PostGIS spatial scaling (10M+ pings)
- Court-admissible evidentiary dossier

OPERATIONAL IMPACT

- **85% Latency Reduction:** Slashes spill origin attribution from 48-72 hours manual plotting to <60s automated pipeline.
- **Evidentiary Traceability:** Generates legally defensible dossiers linking IMO/MMSI numbers with spatio-temporal overlap.
- **Persistent EEZ Surveillance:** Expands autonomous maritime domain awareness across India's 2.37 million km² coastal EEZ.
- **Deterrence Against Dumping:** High attribution probability actively deters commercial ships from nocturnal bilge washing.

ENVIRONMENTAL & ECONOMIC
BENEFITS

- **Ecosystem Preservation:** Protects sensitive mangroves, coral reefs, and fisheries from toxic hydrocarbon contamination.
- **Cost Recovery Enforcement:** Enables port authorities and Coast Guard to recover million-dollar cleanup costs under Polluter Pays.
- **Inter-Agency Compatibility:** Standard GeoJSON and RESTful APIs integrate with Indian Coast Guard (ICG) MRCC systems.
- **Operational Scalability:** Asynchronous multi-queue processing handles nationwide satellite passes without extra personnel.

OPERATIONAL USER SCENARIO (INDIAN MARITIME DOMAIN)

Target User & Deployment Environment

Indian Coast Guard Maritime Rescue Coordination Centre (MRCC) & Directorate General of Shipping operating in national coastal surveillance.

Surveillance Incident

A Sentinel-1 SAR acquisition over the Arabian Sea tanker corridor (Bombay High offshore zone) detects a 4.2 km² dark oil slick.

Autonomous Attribution Workflow

1. U-Net extracts slick polygon boundary and computes centroid (19.42°N, 71.85°E).
2. Lagrangian hindcast steps 16 hours backward against current and wind, locating origin 18.4 nm upstream.
3. Vectorized AIS engine screens 34 commercial vessels in the spatio-temporal corridor.

Forensic Finding

Flags Crude Carrier (MMSI: 413289000) with a 0.94 composite suspect score, identifying a 42-minute AIS transponder blackout gap and a 6.5-knot speed drop at the reconstructed spill release point.

Enforcement Outcome

Evidentiary dossier dispatched to Coast Guard interceptor vessel in under 90 seconds, securing legal prosecution under Indian Merchant Shipping Act 1958.



Synthetic Aperture Radar (SAR) Oil Detection

- European Space Agency (ESA) Sentinel-1 C-Band SAR User Guide & Copernicus Open Access Hub.
- CleanSeaNet Operational Satellite Monitoring Methodology, European Maritime Safety Agency (EMSA).



Deep Learning Semantic Segmentation

- Ronneberger et al., 'U-Net: Convolutional Networks for Biomedical Image Segmentation', MICCAI.
- Solberg et al., 'Algorithms for Oil Spill Detection in SAR Imagery', IEEE Trans. Geosci. Remote Sensing.



Hydrodynamic & Ocean Drift Modeling

- NOAA GNOME (General NOAA Operational Modeling Environment) Technical Suite.
- Dagestad et al., 'OpenDrift: An open-source trajectory framework for ocean drift modeling', GMD.



Meteorological & Current Forcing Datasets

- Copernicus Marine Environment Monitoring Service (CMEMS Global Ocean Current Analysis).
- European Centre for Medium-Range Weather Forecasts (ECMWF) ERA5 Atmospheric Wind Vectors.



Automatic Identification System (AIS) Standards

- International Telecommunication Union (ITU-R M.1371) Technical characteristics for AIS.
- Bureau of Ocean Energy Management (BOEM) & NOAA, MarineCadastre.gov National AIS Database.



Maritime Environmental Law & Regulatory Frameworks

- IMO MARPOL 73/78 Annex I: Regulations for the Prevention of Pollution by Oil from Ships.
- Government of India, Merchant Shipping Act 1958 (Part XIA: Prevention of Pollution of Sea by Oil).